



Image Processing CSE381

Project Phase 1

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project.py



processed_artifacts.zip

1. Introduction

The objective of Milestone 1 was to design a robust image processing pipeline capable of preparing raw images of square-tile puzzles for future assembly. A key feature of our proposed system is its ability to **automatically detect the puzzle grid dimensions** (classifying inputs as 2x2, 4x4, or 8x8) without user intervention. The pipeline addresses challenges such as uneven lighting, sensor noise, and automated segmentation. This report details the "UltimatePipeline" approach.

2. Methodology & Techniques

Our pipeline processes images in four distinct stages. Each technique was selected to maximize feature clarity, which is critical for the pixel-matching tasks in Milestone 2.

2.1. Adaptive Enhancement (CLAHE in LAB Color Space)

Technique: The input BGR image is first converted to the **CIELAB (LAB)** color space. We extract the **L-channel (Lightness)** and apply **Contrast Limited Adaptive Histogram Equalization (CLAHE)** with a clip limit of 2.0 and an 8x8 grid size. The enhanced L-channel is then merged back with the original A and B channels.

Justification: We rejected standard Histogram Equalization because it •
operates globally; a single bright reflection (glare) on a puzzle piece would darken the entire image. CLAHE operates on local tiles, enhancing the local contrast of puzzle borders even in shadowed or overexposed regions. By processing only the L-channel, we strictly preserve the original color information.

2.2. Edge-Preserving Noise Attenuation (Bilateral Filter)

Technique: A **Bilateral Filter** is applied to the contrast-enhanced image.

Justification: Standard Gaussian blurring was deemed unsuitable because it •
smooths edges alongside noise. The Bilateral Filter is non-linear; it averages pixels based on spatial closeness *and* photometric similarity. This effectively removes high-frequency sensor noise (grain) while keeping the straight grid lines sharp, ensuring that the segmentation step relies on accurate cut coordinates.

2.3. Sharpening and Edge Extraction

Technique: We apply a Laplacian-based sharpening kernel ($\begin{bmatrix} -1 & 5 & -1 \end{bmatrix}$...) followed by **Canny Edge Detection** (thresholds 100/200).

Justification: The sharpening kernel creates a "halo" effect around high-contrast transitions, making the Canny detector more reliable. The resulting binary edge map provides a means to verify that the grid cuts align with actual image features. •

2.4. Automatic Grid Size Detection (2x2 / 4x4 / 8x8)

Technique: To segment the pieces, we calculate the sum of absolute gradients along the X and Y axes (Projection Profiles). The system analyzes the normalized profile peaks at specific intervals to mathematically determine the grid density.

Justification: The system is fully autonomous. It tests the projection strength •
for an 8x8 grid first; if the score is low, it falls back to check for a 4x4 grid, and finally defaults to 2x2. This allows the code to **automatically differentiate between 2x2, 4x4, and 8x8 puzzles** and apply the correct slicing logic without the user needing to specify the difficulty level manually.

3. Visual Results

Below is the output of the pipeline showing a sample raw input, the enhanced intermediate stage, and the final extracted contours.



4. Suitability for Assembly (Milestone 2)

The artifacts produced by this pipeline are optimal for the subsequent assembly task. The code successfully generates standardized, noise-free tiles that are ready for immediate use:

Geometric Precision: By utilizing projection profiles, the system guarantees •
that every extracted tile (tile_0_0.png) is a perfect rectangle with no jagged edges. This eliminates the need for geometric post-processing during assembly.

Photometric Consistency: The CLAHE normalization ensures that lighting is •
consistent across all tiles. This effectively standardizes the inputs, ensuring that simple pixel-difference metrics (like SSD) will perform accurately during the matching phase without further adjustment.

Complete Data Separation: The pipeline successfully decouples visual data •
(RGB tiles) from structural data (Edge Maps), providing a comprehensive dataset that supports both texture-based and contour-based matching strategies.